

11th grade
Term 2 Midterm Mock (Hard): Confidence Interval Planning
(Solutions)

Contents

Contents

1. (C1, C5, C6) Maintenance Cost Comparison Across Two Rail Depots
2. (C1, C5, C6) Fraud-Review Time Comparison by Team
3. (C1, C5, C6) Dispatch Completion Time in an Expansion Zone
4. (C1, C5, C6) Audit Value Recovery Comparison Across Field Units
5. (C2, C3, C4) Fintech Repayment Delay Supervision Analysis
6. (C2, C3, C4) Border Port Turnaround Analysis for Policy Decisions
7. (C2, C3, C4) Social Protection Processing-Time Channel Comparison
8. (C2, C3, C4) External Fund Manager Return Reporting Analysis
9. (C7) Supplier On-Time Submission Proportion by Region
10. (C7) Digital Tax Filing Completion by Business Subgroup
11. (C7) Defective Shipment Proportion Across Two Quarterly Audits
12. (C7) Renewable Energy Adoption Proportion Across Two Districts

Evaluated criteria

- C1: Compute the sample mean and sample standard deviation.
- C2: Distinguish between sample statistics and population parameters.
- C3: Explain why interval estimation is preferred over a single point estimate.
- C4: Interpret the meaning of a X% confidence interval in context.
- C5: Construct a X% confidence interval using known population standard deviation.
- C6: Establish the sample size so the error will not exceed a specified amount.
- C7: Construct the confidence interval for a proportion.

Problems

P1 1. (C1, C5, C6) Maintenance Cost Comparison Across Two Rail Depots

A regional rail authority compares average monthly maintenance cost per trainset (thousand USD) in two depots after introducing predictive maintenance software. The grouped data report maintenance-cost values together with their frequencies, where each cost value occurs in a certain number of trainsets.

Depot A: The data correspond to 27 trainsets. Each maintenance-cost value below is measured in thousand USD, and the stated frequency indicates how many trainsets had that value. Cost = 72 (thousand USD) occurs in 5 trainsets; cost = 78 (thousand USD) occurs in 7 trainsets; cost = 85 (thousand USD) occurs in 6 trainsets; cost = 92 (thousand USD) occurs in 5 trainsets; and cost = 100 (thousand USD) occurs in 4 trainsets.

Depot B: The data correspond to 28 trainsets. Each maintenance-cost value below is measured in thousand USD, and the stated frequency indicates how many trainsets had that value. Cost = 68 (thousand USD) occurs in 4 trainsets; cost = 75 (thousand USD) occurs in 6 trainsets; cost = 83 (thousand USD) occurs in 8 trainsets; cost = 90 (thousand USD) occurs in 6 trainsets; and cost = 98 (thousand USD) occurs in 4 trainsets.

From long-run accounting records, the population standard deviation is known to be $\sigma = 14$ thousand USD for both depots.

- Calculate the sample mean and sample standard deviation for each depot using the provided grouped data (C1).
- Construct 90%, 95%, and 99% confidence intervals for each depot mean using known σ , and report the margin of error for each interval (C5).
- Compute required sample sizes for target margins of error $E = 3.5$ and $E = 2.8$ at 90%, 95%, and 99%, and determine additional observations needed for each depot (C6).

C1

Depot A

$$\sum fx = 72(5) + 78(7) + 85(6) + 92(5) + 100(4) = 2276, \quad \bar{x}_A = \frac{2276}{27} = 84.30$$

$$\sum f(x - \bar{x}_A)^2 = 2323.48, \quad s_A^2 = \frac{2323.48}{26} = 89.36, \quad s_A \approx 9.45$$

Depot B

$$\sum fx = 68(4) + 75(6) + 83(8) + 90(6) + 98(4) = 2318, \quad \bar{x}_B = \frac{2318}{28} = 82.79$$

$$\sum f(x - \bar{x}_B)^2 = 2475.64, \quad s_B^2 = \frac{2475.64}{27} = 91.69, \quad s_B \approx 9.58$$

C5

For Depot A, $SE_A = \frac{14}{\sqrt{27}} \approx 2.694$.

$$E_{A,90} = 1.645(2.694) = 4.43, \quad CI_{90} : 84.30 \pm 4.43 = [79.87, 88.73]$$

$$E_{A,95} = 1.96(2.694) = 5.28, \quad CI_{95} : 84.30 \pm 5.28 = [79.02, 89.58]$$

$$E_{A,99} = 2.576(2.694) = 6.94, \quad CI_{99} : 84.30 \pm 6.94 = [77.36, 91.24]$$

For Depot B, $SE_B = \frac{14}{\sqrt{28}} \approx 2.646$.

$$E_{B,90} = 1.645(2.646) = 4.35, \quad CI_{90} : 82.79 \pm 4.35 = [78.43, 87.14]$$

$$E_{B,95} = 1.96(2.646) = 5.19, \quad CI_{95} : 82.79 \pm 5.19 = [77.60, 87.97]$$

$$E_{B,99} = 2.576(2.646) = 6.82, \quad CI_{99} : 82.79 \pm 6.82 = [75.97, 89.60]$$

C6

Use $n = \left(\frac{z^* \sigma}{E}\right)^2$ and round up. For $E = 3.5$:

$$n_{90} = \left(\frac{1.645 \cdot 14}{3.5}\right)^2 = 43.32 \Rightarrow 44, n_{95} = 61.47 \Rightarrow 62, n_{99} = 106.15 \Rightarrow 107$$

Additional observations: Depot A ($n = 27$): 17, 35, 80; Depot B ($n = 28$): 16, 34, 79. For $E = 2.8$:

$$n_{90} = 67.68 \Rightarrow 68, n_{95} = 96.05 \Rightarrow 97, n_{99} = 165.86 \Rightarrow 166$$

Additional observations: Depot A: 41, 70, 139; Depot B: 40, 69, 138.

P2 2. (C1, C5, C6) Fraud-Review Time Comparison by Team

A digital payments platform tracks average fraud-review handling time (minutes) for two independent teams. The grouped data are reported using class intervals, and the frequency for each interval indicates how many cases fall within that time range.

Team A: The data correspond to 38 cases. Each interval below is measured in minutes, and the stated frequency is the number of cases in that interval. 42–46 minutes corresponds to 6 cases; 47–51 minutes corresponds to 9 cases; 52–56 minutes corresponds to 10 cases; 57–61 minutes corresponds to 8 cases; and 62–66 minutes corresponds to 5 cases.

Team B: The data correspond to 40 cases. Each interval below is measured in minutes, and the stated frequency is the number of cases in that interval. 40–44 minutes corresponds to 5 cases; 45–49 minutes corresponds to 8 cases; 50–54 minutes corresponds to 11 cases; 55–59 minutes corresponds to 9 cases; and 60–64 minutes corresponds to 7 cases.

Assume known population standard deviation $\sigma = 6.5$ minutes.

- (a) Calculate the sample mean and sample standard deviation for each team using the provided grouped data (C1).
- (b) Construct 90%, 95%, and 99% confidence intervals for each team mean using known σ , and report the margin of error for each interval (C5).
- (c) Compute required sample sizes for $E = 1.2$ and $E = 0.9$ at 90%, 95%, and 99%, and determine additional cases needed for each team (C6).

C1

Use class midpoints. For Team A, midpoints are 44,49,54,59,64.

$$\sum fm = 44(6) + 49(9) + 54(10) + 59(8) + 64(5) = 2037, \quad \bar{x}_A = \frac{2037}{38} = 53.61$$

$$\sum f(m - \bar{x}_A)^2 = 1520.86, \quad s_A^2 = \frac{1520.86}{37} = 41.10, \quad s_A \approx 6.41$$

For Team B, midpoints are 42,47,52,57,62.

$$\sum fm = 42(5) + 47(8) + 52(11) + 57(9) + 62(7) = 2105, \quad \bar{x}_B = \frac{2105}{40} = 52.63$$

$$\sum f(m - \bar{x}_B)^2 = 1610.94, \quad s_B^2 = \frac{1610.94}{39} = 41.31, \quad s_B \approx 6.42$$

C5

$$SE_A = \frac{6.5}{\sqrt{38}} = 1.055, \quad E_{A,90} = 1.645(1.055) = 1.73, \quad E_{A,95} = 1.96(1.055) = 2.07, \quad E_{A,99} = 2.576(1.055) = 2.72$$

$$CI_{A,90} = [51.87, 55.34], \quad CI_{A,95} = [51.54, 55.67], \quad CI_{A,99} = [50.89, 56.32]$$

$$SE_B = \frac{6.5}{\sqrt{40}} = 1.028, \quad E_{B,90} = 1.69, \quad E_{B,95} = 2.01, \quad E_{B,99} = 2.65$$

$$CI_{B,90} = [50.93, 54.32], \quad CI_{B,95} = [50.61, 54.64], \quad CI_{B,99} = [49.98, 55.27]$$

C6

For $E = 1.2$:

$$n_{90} = \left(\frac{1.645 \cdot 6.5}{1.2} \right)^2 = 79.36 \Rightarrow 80, n_{95} = 112.69 \Rightarrow 113, n_{99} = 194.65 \Rightarrow 195$$

Additional cases needed: Team A ($n = 38$): 42, 75, 157; Team B ($n = 40$): 40, 73, 155. For $E = 0.9$:

$$n_{90} = 141.08 \Rightarrow 142, n_{95} = 200.34 \Rightarrow 201, n_{99} = 346.04 \Rightarrow 347$$

Additional cases needed: Team A: 104, 163, 309; Team B: 102, 161, 307.

P3 3. (C1, C5, C6) Dispatch Completion Time in an Expansion Zone

An emergency logistics center records warehouse dispatch completion times (hours) for one expansion zone. A random sample of 18 dispatches is: 31, 29, 34, 28, 33, 35, 30, 32, 36, 27, 34, 31, 29, 33, 30, 37, 28, 32. Known population standard deviation is $\sigma = 5.8$ hours.

- Calculate the sample mean and sample standard deviation for the dispatch-time sample (C1).
- Construct 90%, 95%, and 99% confidence intervals for the population mean completion time using known σ , and report the margin of error for each interval (C5).
- Compute required sample sizes for target errors $E = 2.0$ and $E = 1.5$ at 90%, 95%, and 99%, and determine additional dispatches needed (C6).

C1

$$\bar{x} = \frac{569}{18} = 31.61$$

$$\sum (x_i - \bar{x})^2 = 142.28, \quad s^2 = \frac{142.28}{17} = 8.37, \quad s = 2.89$$

C5

$$SE = \frac{5.8}{\sqrt{18}} = 1.367$$

$$E_{90} = 1.645(1.367) = 2.25, \quad CI_{90} = 31.61 \pm 2.25 = [29.36, 33.86]$$

$$E_{95} = 1.96(1.367) = 2.68, \quad CI_{95} = 31.61 \pm 2.68 = [28.93, 34.29]$$

$$E_{99} = 2.576(1.367) = 3.52, \quad CI_{99} = 31.61 \pm 3.52 = [28.09, 35.13]$$

C6

For $E = 2.0$:

$$n_{90} = \left(\frac{1.645 \cdot 5.8}{2.0} \right)^2 = 22.75 \Rightarrow 23, n_{95} = 32.31 \Rightarrow 33, n_{99} = 55.83 \Rightarrow 56$$

Additional dispatches from current $n = 18$: 5, 15, 38. For $E = 1.5$:

$$n_{90} = 40.44 \Rightarrow 41, n_{95} = 57.43 \Rightarrow 58, n_{99} = 99.26 \Rightarrow 100$$

Additional dispatches: 23, 40, 82.

P4 4. (C1, C5, C6) Audit Value Recovery Comparison Across Field Units

A tax administration compares average daily audit value recovered (thousand USD) by two independent field units.

Unit A (15 days): 112, 108, 115, 121, 117, 109, 114, 118, 116, 113, 120, 111, 119, 107, 122.

Unit B (18 days): 105, 109, 111, 114, 108, 112, 116, 110, 113, 107, 115, 118, 106, 109, 117, 111, 114, 108.

Assume known $\sigma = 9.2$ thousand USD in both units.

- Calculate the sample mean and sample standard deviation for each unit using the provided daily values (C1).
- Construct 90%, 95%, and 99% confidence intervals for each unit mean using known σ , and report the margin of error for each interval (C5).
- Compute required sample sizes for $E = 3.0$ and $E = 2.2$ at 90%, 95%, and 99%, and determine additional observations needed for each unit (C6).

C1

$$\bar{x}_A = \frac{1722}{15} = 114.80, \quad s_A \approx 4.77$$
$$\bar{x}_B = \frac{2003}{18} = 111.28, \quad s_B \approx 3.85$$

C5

For Unit A, $SE_A = \frac{9.2}{\sqrt{15}} = 2.375$.

$$E_{A,90} = 3.91, \quad CI_{A,90} = [110.89, 118.71]$$

$$E_{A,95} = 4.66, \quad CI_{A,95} = [110.14, 119.46]$$

$$E_{A,99} = 6.12, \quad CI_{A,99} = [108.68, 120.92]$$

For Unit B, $SE_B = \frac{9.2}{\sqrt{18}} = 2.168$.

$$E_{B,90} = 3.57, \quad CI_{B,90} = [107.71, 114.84]$$

$$E_{B,95} = 4.25, \quad CI_{B,95} = [107.03, 115.53]$$

$$E_{B,99} = 5.59, \quad CI_{B,99} = [105.69, 116.86]$$

C6

For $E = 3.0$:

$$n_{90} = 25.44 \Rightarrow 26, \quad n_{95} = 36.14 \Rightarrow 37, \quad n_{99} = 62.44 \Rightarrow 63$$

Additional days needed: Unit A ($n = 15$): 11, 22, 48; Unit B ($n = 18$): 8, 19, 45. For $E = 2.2$:

$$n_{90} = 47.28 \Rightarrow 48, \quad n_{95} = 67.17 \Rightarrow 68, \quad n_{99} = 116.11 \Rightarrow 117$$

Additional days needed: Unit A: 33, 53, 102; Unit B: 30, 50, 99.

P5 5. (C2, C3, C4) Fintech Repayment Delay Supervision Analysis

A central bank supervision office evaluates a fintech lender. One team reports from a recent compliance review: average repayment delay was 4.7 days with spread 2.1 days, and a 95% confidence interval for the mean delay was [4.1, 5.3]. A second team cites long-run portfolio values from the full national borrower base: true average delay = 4.9 days and true standard deviation = 2.4 days. The policy division also references a strategic threshold: “acceptable mean delay below 5.0 days.”

Explicit analysis tasks: 1) Classify every numerical quantity above as a statistic from observed data, a fixed characteristic of the full borrower base, or a policy benchmark, and justify (C2). 2) Using the reported interval for mean repayment delay, answer all of the following (C3): (i) state the point estimate for the mean repayment delay; (ii) explain, in this supervision context, why reporting only that point estimate is insufficient compared to reporting the interval; (iii) compute the interval width and explain what it says about precision; (iv) propose two concrete changes to the data-collection or measurement design that would narrow the interval, and justify briefly. 3) State which mean values are ruled out/plausible under [4.1, 5.3] and explain “95% confidence” in repeated-trial language (C4).

C2

Observed-data statistics (from the recent compliance review). The sample mean repayment delay is $\bar{x} = 4.7$ days. The sample spread is $s = 2.1$ days. The reported 95% confidence interval for the population mean is [4.1, 5.3], so 4.1 and 5.3 are interval bounds computed from sample data.

Full-population characteristics (long-run national borrower base). The true population mean is $\mu = 4.9$ days. The true population standard deviation is $\sigma = 2.4$ days. These are parameters because they describe the full borrower population, not just one reviewed sample.

Policy benchmark (decision threshold). The quantity “acceptable mean delay below 5.0 days” is a policy cutoff, not a sample statistic and not a population parameter estimated from this review.

C3

(i) Point estimate for the mean delay. From the report, the point estimate is

$$\hat{\mu} = \bar{x} = 4.7 \text{ days.}$$

(ii) Why the point estimate alone is not enough in supervision decisions. The single value 4.7 days gives only a center. It does not communicate estimation uncertainty. The interval [4.1, 5.3] adds this uncertainty by showing a range of plausible population means, which is essential when the policy threshold is 5.0 days.

(iii) Interval width and precision.

$$\text{Width} = 5.3 - 4.1 = 1.2 \text{ days.}$$

A width of 1.2 days indicates moderate precision; a smaller width would indicate a more precise estimate of the true mean delay.

(iv) Two design changes that would narrow the interval. Increase sample size in the compliance review, which lowers the standard error. Improve measurement consistency (for example, one standardized timestamp protocol), which can reduce variability in recorded delays.

C4

The reported 95% confidence interval is

$$\mu \in [4.1, 5.3] \text{ days.}$$

So population-mean values between 4.1 and 5.3 days are plausible under this evidence, while values below 4.1 or above 5.3 are ruled out by this interval report.

The 95% confidence statement is about the method: if the same sampling and interval procedure were repeated many times under similar conditions, about 95% of those intervals would capture the true population mean delay.

P6 6. (C2, C3, C4) Border Port Turnaround Analysis for Policy Decisions

A national logistics ministry monitors truck turnaround time at border ports. A pilot review from Port North gives mean 11.8 hours and 90% CI [10.9, 12.7]. A separate audit from Port South gives mean 12.4 hours and 90% CI [11.6, 13.2]. Historic all-port values for the old regime were true average turnaround = 12.1 hours and true standard deviation = 3.0 hours. A policy KPI requires maintaining average turnaround at or below 12.0 hours.

Explicit analysis tasks: 1) Classify observed-data quantities, full-system characteristics, and policy quantities (C2). 2) Using the reported intervals for mean turnaround time, answer all of the following (C3): (i) state the point estimate for Port North and the point estimate for Port South; (ii) explain, in this policy-decision context, why reporting only those point estimates is insufficient compared to reporting intervals; (iii) compare precision by computing the widths of the North and South intervals and identify which report is more precise; (iv) propose two concrete changes to the data-collection or measurement design that would narrow the interval(s), and justify briefly. 3) For each port, state ruled-out/plausible means and explain operational meaning of 90% confidence (C4).

C2

Observed-data statistics (pilot and audit reports). Port North sample mean: $\bar{x}_N = 11.8$ hours, with reported 90% interval [10.9, 12.7]. Port South sample mean: $\bar{x}_S = 12.4$ hours, with reported 90% interval [11.6, 13.2]. These means and interval endpoints come from observed samples.

Full-system characteristics (historic old regime). True all-port average turnaround: $\mu = 12.1$ hours. True all-port standard deviation: $\sigma = 3.0$ hours. These are population parameters for the historic system.

Policy quantity (KPI threshold). The requirement “average turnaround at or below 12.0 hours” is a policy benchmark used for evaluation.

C3

(i) Point estimates for each port. Port North: $\hat{\mu}_N = 11.8$ hours. Port South: $\hat{\mu}_S = 12.4$ hours.

(ii) Why point estimates alone are insufficient. Point estimates provide center values but not uncertainty. Because decisions depend on the KPI cutoff of 12.0 hours, interval estimates are needed to judge whether each port’s true mean could realistically be below, near, or above that threshold.

(iii) Precision comparison using interval widths.

$$W_N = 12.7 - 10.9 = 1.8, \quad W_S = 13.2 - 11.6 = 1.6$$

Port South is slightly more precise because its interval is narrower.

(iv) Two changes that would narrow interval(s). Increase sample sizes at both ports to reduce standard error. Standardize timing and measurement protocols across ports to reduce process and recording variability.

C4

For Port North, the 90% interval is

$$\mu_N \in [10.9, 12.7].$$

So values inside this range are plausible for the true mean turnaround at Port North, while values outside this range are ruled out by this report.

For Port South, the 90% interval is

$$\mu_S \in [11.6, 13.2].$$

So values inside this range are plausible for the true mean turnaround at Port South, while values outside this range are ruled out by this report.

The 90% confidence interpretation is repeated-trial based: if comparable studies were repeated many times and intervals were built the same way, about 90% of those intervals would contain the true port mean.

P7 7. (C2, C3, C4) Social Protection Processing-Time Channel Comparison

A social protection agency studies benefit-payment processing time across two digital channels. Channel A monitoring report gives mean 6.3 days with CI [5.8, 6.8] at 95%. Channel B audit report gives mean 6.0 days with CI [5.2, 6.8] at 95%. Legacy national values before automation were true average processing time = 6.7 days and true standard deviation = 1.9 days. The ministry also publishes two benchmarks: warning level 6.5 days, excellent level 6.0 days.

Explicit analysis tasks: 1) Categorize all values into observed-data statistics, fixed national characteristics, and policy benchmarks (C2). 2) Using the reported intervals for mean processing time, answer all of the following (C3): (i) state the point estimate for Channel A and the point estimate for Channel B; (ii) explain, in this social-protection context, why reporting only those point estimates is insufficient compared to reporting intervals; (iii) compare precision by computing the widths of the Channel A and Channel B intervals and identify which report is more precise; (iv) propose two concrete changes to the data-collection or measurement design that would narrow the interval(s), and justify briefly. 3) Determine plausible/ruled-out means per channel and explain repeated-trial meaning of 95% confidence (C4).

C2

Observed-data statistics (channel reports). Channel A sample mean is $\bar{x}_A = 6.3$ days with 95% interval [5.8, 6.8]. Channel B sample mean is $\bar{x}_B = 6.0$ days with 95% interval [5.2, 6.8]. These values are computed from observed samples.

Fixed national characteristics (legacy period). Legacy true mean processing time is $\mu = 6.7$ days. Legacy true standard deviation is $\sigma = 1.9$ days. These are population parameters from the pre-automation national system.

Policy benchmarks. Warning level: 6.5 days. Excellent level: 6.0 days. Both are policy targets used for operational evaluation.

C3

(i) Point estimates for each channel. Channel A: $\hat{\mu}_A = 6.3$ days. Channel B: $\hat{\mu}_B = 6.0$ days.

(ii) Why point estimates alone are insufficient. A point estimate gives one best guess but not uncertainty. Intervals show the plausible range for each channel's true mean, which is necessary for safer planning around warning and excellence benchmarks.

(iii) Precision comparison using interval widths.

$$W_A = 6.8 - 5.8 = 1.0, \quad W_B = 6.8 - 5.2 = 1.6$$

Channel A is more precise because its interval is narrower.

(iv) Two changes that would narrow interval(s). Increase sample sizes for both channels. Improve data-quality consistency (for example, uniform timestamp rules and case-classification standards) to reduce variability.

C4

For Channel A, the 95% interval is

$$\mu_A \in [5.8, 6.8].$$

So true-mean values inside this range are plausible, and values outside this range are ruled out by this interval report.

For Channel B, the 95% interval is

$$\mu_B \in [5.2, 6.8].$$

So true-mean values inside this range are plausible, and values outside this range are ruled out by this interval report.

The 95% confidence meaning is procedural: over many repeated comparable samples, about 95% of intervals built this way would contain the true channel mean.

P8 8. (C2, C3, C4) External Fund Manager Return Reporting Analysis

A sovereign wealth monitoring office evaluates monthly return volatility reporting from two external fund managers. Manager Alpha reports a mean monthly return of 1.4% with 95% CI [0.8%, 2.0%]. Manager Beta reports a mean of 1.1% with 95% CI [0.2%, 2.0%]. Historical long-run values from the full investment universe are true average monthly return = 1.3% and true standard deviation = 3.5%. The board uses two policy thresholds: “underperforming” if mean return < 1.0% and “stretch target” if mean return $\geq$ 1.8%.

Explicit analysis tasks: 1) Classify all quantities into observed-data statistics, fixed universe characteristics, and policy thresholds (C2). 2) Using the reported intervals for mean monthly return, answer all of the following (C3): (i) state the point estimate for Manager Alpha and the point estimate for Manager Beta; (ii) explain, in this strategic-allocation context, why reporting only those point estimates is insufficient compared to reporting intervals; (iii) compare precision by computing the widths of the Alpha and Beta intervals and identify which report is more precise; (iv) propose two concrete changes to the data-collection or measurement design that would narrow the interval(s), and justify briefly. 3) For each manager, state ruled-out and plausible values and explain repeated-trial meaning of 95% confidence (C4).

C2

Observed-data statistics (manager reports). Manager Alpha sample mean monthly return: $\bar{x}_\alpha = 1.4\%$ with 95% interval [0.8%, 2.0%]. Manager Beta sample mean monthly return: $\bar{x}_\beta = 1.1\%$ with 95% interval [0.2%, 2.0%]. These means and bounds are statistics obtained from sample reporting windows.

Fixed universe characteristics (full investment universe). True long-run mean monthly return: $\mu = 1.3\%$. True long-run standard deviation: $\sigma = 3.5\%$. These are population parameters.

Policy thresholds (board rules). Underperforming threshold: 1.0%. Stretch-target threshold: 1.8%. These are decision benchmarks, not estimated sample statistics.

C3

(i) Point estimates for each manager. Manager Alpha: $\hat{\mu}_\alpha = 1.4\%$. Manager Beta: $\hat{\mu}_\beta = 1.1\%$.

(ii) Why point estimates alone are insufficient. Point estimates summarize central performance but hide uncertainty. Intervals are needed to judge how plausible underperformance (< 1.0%) or stretch-target performance ($\geq$ 1.8%) is for each manager.

(iii) Precision comparison using interval widths.

$$W_\alpha = 2.0 - 0.8 = 1.2\%, \quad W_\beta = 2.0 - 0.2 = 1.8\%$$

Manager Alpha is more precise because the interval is narrower.

(iv) Two changes that would narrow interval(s). Use larger sample windows (more observations). Standardize reporting periods and return definitions to reduce measurement inconsistency and variability.

C4

For Manager Alpha, the 95% interval is

$$\mu_\alpha \in [0.8\%, 2.0\%].$$

So values within this band are plausible for Alpha's true mean monthly return, while values outside the band are ruled out by this interval report.

For Manager Beta, the 95% interval is

$$\mu_{\beta} \in [0.2\%, 2.0\%].$$

So values within this band are plausible for Beta's true mean monthly return, while values outside the band are ruled out by this interval report.

The repeated-trial interpretation of 95% confidence is that, under the same sampling procedure repeated many times, about 95% of constructed intervals would capture the true manager mean return.

P9 9. (C7) Supplier On-Time Submission Proportion by Region

A public procurement platform reports supplier on-time document submission rates for two regions. Region East sample: 420 suppliers with 246 on-time. Region West sample: 380 suppliers with 186 on-time.

- (a) Construct 90%, 95%, and 99% confidence intervals for each region's population proportion, and report the margin of error for each interval (C7).

C7**Region East**

$$\hat{p}_E = \frac{246}{420} = 0.5857, \quad SE_E = \sqrt{\frac{0.5857(0.4143)}{420}} = 0.0240$$

$$E_{90} = 1.645(0.0240) = 0.0395, \quad CI_{90} = [0.5462, 0.6253]$$

$$E_{95} = 1.96(0.0240) = 0.0471, \quad CI_{95} = [0.5386, 0.6328]$$

$$E_{99} = 2.576(0.0240) = 0.0619, \quad CI_{99} = [0.5238, 0.6476]$$

Region West

$$\hat{p}_W = \frac{186}{380} = 0.4895, \quad SE_W = \sqrt{\frac{0.4895(0.5105)}{380}} = 0.0256$$

$$E_{90} = 1.645(0.0256) = 0.0422, \quad CI_{90} = [0.4473, 0.5317]$$

$$E_{95} = 1.96(0.0256) = 0.0503, \quad CI_{95} = [0.4392, 0.5397]$$

$$E_{99} = 2.576(0.0256) = 0.0661, \quad CI_{99} = [0.4234, 0.5555]$$

P10 10. (C7) Digital Tax Filing Completion by Business Subgroup

A municipal e-governance program evaluates digital tax filing completion by business size subgroup in one city. Small firms: $n = 250$, completed $x = 142$. Medium firms: $n = 200$, completed $x = 96$. Large firms: $n = 150$, completed $x = 57$.

- (a) Construct 90% and 95% confidence intervals for each subgroup proportion, and report the margin of error for each interval (C7).

C7**Small firms**

$$\begin{aligned}\hat{p}_S &= \frac{142}{250} = 0.568, & SE_S &= \sqrt{\frac{0.568(0.432)}{250}} = 0.0313 \\ E_{90} &= 1.645(0.0313) = 0.0515, & CI_{90} &= [0.5165, 0.6195] \\ E_{95} &= 1.96(0.0313) = 0.0614, & CI_{95} &= [0.5066, 0.6294]\end{aligned}$$

Medium firms

$$\begin{aligned}\hat{p}_M &= \frac{96}{200} = 0.480, & SE_M &= \sqrt{\frac{0.48(0.52)}{200}} = 0.0353 \\ E_{90} &= 1.645(0.0353) = 0.0581, & CI_{90} &= [0.4219, 0.5381] \\ E_{95} &= 1.96(0.0353) = 0.0692, & CI_{95} &= [0.4108, 0.5492]\end{aligned}$$

Large firms

$$\begin{aligned}\hat{p}_L &= \frac{57}{150} = 0.380, & SE_L &= \sqrt{\frac{0.38(0.62)}{150}} = 0.0396 \\ E_{90} &= 1.645(0.0396) = 0.0652, & CI_{90} &= [0.3148, 0.4452] \\ E_{95} &= 1.96(0.0396) = 0.0777, & CI_{95} &= [0.3023, 0.4577]\end{aligned}$$

P11 11. (C7) Defective Shipment Proportion Across Two Quarterly Audits

A pharmaceutical procurement unit tracks defective shipment proportion in two independent quarterly audits. Quarter 1: $n_1 = 520$, defectives $x_1 = 57$. Quarter 2: $n_2 = 610$, defectives $x_2 = 52$.

- (a) Construct 95% and 99% confidence intervals for each quarter proportion, and report the margin of error for each interval (C7).

C7**Quarter 1**

$$\begin{aligned}\hat{p}_1 &= \frac{57}{520} = 0.1096, & SE_1 &= \sqrt{\frac{0.1096(0.8904)}{520}} = 0.0137 \\ E_{95} &= 1.96(0.0137) = 0.0269, & CI_{95} &= [0.0828, 0.1365] \\ E_{99} &= 2.576(0.0137) = 0.0353, & CI_{99} &= [0.0743, 0.1449]\end{aligned}$$

Quarter 2

$$\begin{aligned}\hat{p}_2 &= \frac{52}{610} = 0.0852, & SE_2 &= \sqrt{\frac{0.0852(0.9148)}{610}} = 0.0113 \\ E_{95} &= 1.96(0.0113) = 0.0222, & CI_{95} &= [0.0631, 0.1074] \\ E_{99} &= 2.576(0.0113) = 0.0291, & CI_{99} &= [0.0561, 0.1144]\end{aligned}$$

P12 12. (C7) Renewable Energy Adoption Proportion Across Two Districts

A national energy agency studies the proportion of households that adopted rooftop solar systems in two districts during the same year. District North: $n_1 = 480$, adopters $x_1 = 168$. District South: $n_2 = 540$, adopters $x_2 = 221$.

- (a) Construct 90%, 95%, and 99% confidence intervals for each district proportion, and report the margin of error for each interval (C7).

C7**District North**

$$\begin{aligned}\hat{p}_N &= \frac{168}{480} = 0.3500, & SE_N &= \sqrt{\frac{0.35(0.65)}{480}} = 0.0218 \\ E_{90} &= 1.645(0.0218) = 0.0358, & CI_{90} &= [0.3142, 0.3858] \\ E_{95} &= 1.96(0.0218) = 0.0427, & CI_{95} &= [0.3073, 0.3927] \\ E_{99} &= 2.576(0.0218) = 0.0561, & CI_{99} &= [0.2939, 0.4061]\end{aligned}$$

District South

$$\begin{aligned}\hat{p}_S &= \frac{221}{540} = 0.4093, & SE_S &= \sqrt{\frac{0.4093(0.5907)}{540}} = 0.0212 \\ E_{90} &= 1.645(0.0212) = 0.0348, & CI_{90} &= [0.3745, 0.4441] \\ E_{95} &= 1.96(0.0212) = 0.0415, & CI_{95} &= [0.3678, 0.4507] \\ E_{99} &= 2.576(0.0212) = 0.0545, & CI_{99} &= [0.3548, 0.4638]\end{aligned}$$